

Chapter P2: Sustainable energy

Overview

Energy supply is one of the major issues that society must address in the immediate future.

Citizens are faced with complex choices and a variety of messages from energy supply companies, environmental groups, the media, scientists and politicians. Some maintain that renewable resources are capable of meeting our future needs, some advocate nuclear power, and some argue that drastic lifestyle changes are required. Decisions about energy use, whether at a personal or a national level, need to be informed by a quantitative understanding of the

situation, and this is an underlying theme of the chapter.

Topic P2.1 quantifies the energy used by electrical devices introduces calculations of efficiency and considers ways of reducing dissipation in a variety of contexts. In Topic P2.2 national data on energy sources introduces a study of electricity generation and distribution; nuclear power generation, the burning of fossil fuels and renewable resources are compared and contrasted. Electrical safety in the home and a review the energy choices available to individuals, organisations and society are also included.

Learning about energy before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- have compared energy uses and costs in domestic contexts, including calculations using a variety of units
- have considered a variety of processes that involve transferring energy, including heating, changing motion, burning fuels and changing position in a field.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about sustainable energy at GCSE (9–1)

P2.1 How much energy do we use?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Energy is considered as being stored in a limited number of ways: chemical, nuclear, kinetic, gravitational, elastic, thermal, electrostatic and electromagnetic and can be transferred from one to another by processes called working and heating. Electricity is a convenient way to transfer energy from source to the consumer because it is easily transmitted over distances and can be used to do work in many ways, including heating and driving motors which make things move or to lift weights. When energy is used to do work some energy is usually wasted in doing things other than the intended outcome, it is dissipated into the surroundings, ultimately into inaccessible thermal stores. The power of an appliance or device is a measure of the amount of energy it transfers each second, i.e. the rate at which it transfers energy.	1. describe how energy in chemical stores in batteries, or in fuels at the power station, is transferred by an electric current, doing work on domestic devices, such as motors or heaters
	2. explain, with reference to examples, the relationship between the power ratings for domestic electrical appliances, the time for which they are in use and the changes in stored energy when they are in use
	3. recall and apply the following equation in the context of energy transfers by electrical appliances: energy transferred (J, kWh) = power (W, kW) × time (s, h) M3b, M3c, M3d
	4. describe, with examples, where there are energy transfers in a system, that there is no net change to the total energy of a closed system <i>qualitative only</i>
	5. describe, with examples, system changes, where energy is dissipated, so that it is stored in less useful ways
	6. explain ways of reducing unwanted energy transfer e.g. through lubrication, thermal insulation
	7. describe the effects, on the rate of cooling of a building, of thickness and thermal conductivity of its walls <i>qualitative only</i>

Linked learning opportunities

Practical work

- Compare the power consumption of a variety of devices and relate it to the changes in stored energy.
- Investigate the effects of insulation on the rate of cooling.

Maths

- Calculate the cost of energy supplied by electricity given the power rating, the time and the cost per kWh (1aS2.2).

P2.1 How much energy do we use?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Sankey diagrams are used to show all the energy transfers in a system, including energy dissipated to the surroundings; the data can be used to calculate the efficiency of energy transfers.	8. recall and apply the equation: $\text{efficiency} = \frac{\text{useful energy transferred (J)}}{\text{total energy transferred (J)}}$ to calculate energy efficiency for any energy transfer, and describe ways to increase efficiency M1c
	9. interpret and construct Sankey diagrams to show understanding that energy is conserved M4a

Linked learning opportunities

P2.2 How can electricity be generated?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
The main energy resources that are available to humans are fossil fuels (oil, gas, coal), nuclear fuels, biofuels, wind, hydroelectric, tides and from the Sun. In most power stations generators produce a voltage across a wire by spinning a magnet near the wire. Often an energy source is used to heat water; the steam produced drives a turbine which is coupled to an electrical generator. Other energy sources drive the generator directly. The mains supply to our homes is an alternating voltage, at 50 Hz, 230 volts, but electricity is distributed through the National Grid at much higher voltages to reduce energy losses. Transformers are used to increase the voltage for transmission and then decrease the voltage for domestic use. Most mains appliances are connected by a 3 core cable, containing live, neutral and earth wires. The demand for energy is continually increasing and this raises issues about the availability and sustainability of energy sources and the environmental effects of using these sources. The introduction and development of new energy sources may provide new opportunities but also introduce technological and environmental challenges. The decisions about the energy sources that are used may be different for different people in different contexts (1aS4).	1. describe the main energy resources available for use on Earth (including fossil fuels, nuclear fuel, biofuel, wind, hydroelectricity, the tides and the Sun)
	2. explain the differences between renewable and non-renewable energy resources
	3. compare the ways in which the main energy resources are used to generate electricity M2c
	4. recall that the domestic supply in the UK is a.c., at 50 Hz and about 230 volts and explain the difference between direct and alternating voltage
	5. recall that, in the National Grid, transformers are used to transfer electrical power at high voltages from power stations, to the network and then used again to transfer power at lower voltages in each locality for domestic use
	6. recall the differences in function between the live, neutral and earth mains wires, and the potential differences between these wires; hence explain that a live wire may be dangerous even when a switch in a mains circuit is open, and explain the dangers of providing any connection between the live wire and any earthed object
	7. explain patterns and trends in the use of energy resources in domestic contexts, workplace contexts, and national contexts M2c

Linked learning opportunities

Specification links

- What determines the rate of energy transfer in a circuit? (P3.4)
- What is the process inside a generator? (P3.7)

Practical work

- Investigate factors affecting the output from solar panels and wind turbines.

Maths

- Use ideas about probability in the context of risk.
- Extract and interpret information about electricity generation and energy use presented in a variety of numerical and graphical forms

Ideas about Science

- Discuss the risks and benefits of different ways of generating electricity and why different decisions on the same issue might be appropriate (1aS4.3–4.6, 4.11).